C-F-158c
April 9, 1969
SUPERSEDING
Fed. Spec. C-F-158b
December 10, 1965 and
MIL-B-11836B(GL)
November 13, 1964
(See 6.4)

FEDERAL SPECIFICATION

FEATHERS, LANDFOWL; (WHOLE) AND (CRUSHED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for whole and crushed chicken feathers.

1.2 Classification. The landfowl feathers shall be of the following types as specified (see 6.2):

Type I - Whole landfowl feathers (for use as filling material).
Type II - Whole landfowl feathers (for use in filtration).
Type III - Crushed landfowl feathers (for use as filling material).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276 - Thread, Cotton.
QQ-S-781 - Steel, Strapping, Flat.
CCC-C-429 - Cloth, Osnaburg, Cotton.
CCC-C-432 - Cloth, Sheeting, Cotton, (Unbleached, Bleached, and Dyed).
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-S-760 - Strapping, Nonmetallic, (and Connectors).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 148 - Classification, Identification, and Testing of Feather Filling Material.
Fed. Std. No. 751 - Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

FSC 8320

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PUBLIC HEALTH ASSOCIATION, INC.

Standard Methods for the Examination of Water, Sewage, and Industrial Waste.

(Application for copies should be addressed to the American Public Health Association, Inc., 1790 Broadway, New York, N.Y. 10019.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), before production is commenced, a one-pound sample of the finished product shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all the requirements of this specification.

3.2 Materials. The feathers shall be from domesticated chickens and shall be new (not previously used or reclaimed). The feathers shall be of such composition and quality to yield finished processed feathers meeting the requirements of this specification.

3.2.1 Preservation of raw feathers. Not more than 24 hours shall be permitted to elapse between time of plucking and beginning of processing as described in 3.3. Feathers which are not to be processed within 12 hours after plucking shall be preserved by steeping in an acid bath containing approximately 10 ounces of a commercial laundry sour of the fluoride type (sodium acid fluoride, sodium silicofluoride, or combination thereof) in each 100 gallons of water. Feathers which have been held by acid steeping shall be rinsed with clean water before proceeding with the processing described in 3.3.

3.3 Processing.

3.3.1 Cleaning. The feathers shall be washed blood-free with the aid of a blood-solubilizing agent, scoured with detergent, and rinsed. For types I and III feathers only, the final rinse shall contain a commercial laundry sour of the fluoride type (see 3.2.1). The feathers shall then be dried at sufficiently high temperature to effect sterilizing conditions (see 6.3).

3.3.2 Crushing (type III). Type III feathers shall be processed by a crushing or curling machine to alter the original flat configuration of the feathers without removing the quill.

3.4 Finished requirements.

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3.4.2 Oxygen number. The finished product shall show an oxygen number not greater than the following when tested as specified in 4.5.

Types I and II - 10.0 maximum
Type III - 12.0 maximum

3.4.3 Solvent soluble material. The solvent soluble material in the finished product shall not exceed 2.5 percent when tested as specified in 4.5.

3.4.4 Brittleness (types I and II). Types I and II finished products shall show no more brittleness than 8.0 percent failures when tested as specified in 4.5.

3.4.5 Filling power (type I). The filling power of the type I finished product shall be not less than 2.6 centimeters when tested as specified in 4.5.

3.4.6 Turbidity (type II). The turbidity of the type II feathers shall be not greater than that shown by 130 parts per million of silica in water when tested as specified in 4.5.

3.4.7 Composition. The finished product shall conform to the applicable requirements for composition in table I when tested as specified in 4.5. Classification of components shall be in accordance with the applicable definitions in Fed. Std. No. 148.

TABLE I. Composition (percent by weight)

	Type I	Type II	Type III
Whole feathers longer than 3-1/2 inches	10.0 max	—	--
Whole feathers 3-1/2 inches or less in length	72.0 min	—	--
Whole feathers 2 to 3-1/2 inches in length	—	80.0 min	—
Whole feathers shorter than 2 inches, and longer than 3-1/2 inches	—	13.0 max	--
Whole feathers (including quill feathers 4 inches or less)	--	--	Remainder
Crushed feathers and feather fiber	--	--	75.0 min
Quill feathers over 4 inches	--	—	3.0 max
Quill feathers over 3-1/2 inches	--	0	--
Quill feathers (regardless of length)	1.0 max	1/	1/
Damaged feathers (and stripped feathers)	15.0 max	5.0 max	--
Residual matter (including stripped quill shafts)	--	—	5.0 max
Residual matter (and feather fiber)	2.0 max	2.0 max	--
Feather fiber	--	--	40.0 max

1/ See requirements for quill feathers over 3-1/2 inches and over 4 inches.

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3.5 Workmanship. The finished product shall be uniformly processed, of uniform composition, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. The preproduction sample submitted in accordance with 3.1 shall be examined and tested to determine compliance with the requirements of section 3.

4.3 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4 In-process inspection. Inspection shall be conducted to see that the processing is in accordance with the specified requirements. The supplier shall certify conformance to the following requirements:

(a) Use of new chicken feathers put into processing within 24 hours from time of plucking (see 3.2 and 3.2.1).
(b) Use of specified acid steep if feathers are not put into processing within 12 hours of plucking (see 3.2.1).
(c) Conduct of processing as specified (see 3.3).

4.5 Testing of the end item. Testing of the finished product shall be performed in accordance with the methods specified in table II. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The test methods indicated by number shall be performed as specified in Fed. Std. No. 148. The sample unit for testing shall be a thoroughly mixed 24-ounce composite of the finished product. Not more than 6 ounces shall be taken from any one 50-pound bag (see 5.1.1). For larger bags, samplings shall be from top, center, bottom and opposite sides of the bag, with not more than 12 ounces from any one bag. Care shall be taken in mixing to prevent damaging or compressing of the material. The lot size shall be expressed in terms of pounds. The sample size (number of sample units) and the acceptance number shall be in accordance with the following:

Lot size (pounds)	Sample size (units)	Maximum acceptance number of test failures
3,200 and less	2	0
3,201 up to and including 10,000	3	0
10,001 up to and including 35,000	5	0
35,001 and over	8	1

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TABLE II. Test methods 1/

Characteristic	Requirement reference	Test method
Material identification	3.2	2/
Acidity (pH)	3.4.1	8
Oxygen number	3.4.2	4
Solvent soluble material	3.4.3	5
Brittleness (types I and II)	3.4.4	3
Composition	3.4.7	2
Filling power (type I)	3.4.5	1
Turbidity (type II)	3.4.6	4.5.1

1/ Testing may be performed under prevailing conditions except in settlement of dispute in which case standard conditions as defined in Fed. Std. No. 148 shall apply.

2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.5.1 Determination of turbidity (type II).

4.5.1.1 Test specimen. The test specimen shall be a suitable portion of the filtrate obtained as specified for determination of oxygen number (Test Method 4).

4.5.1.2 Procedure. The transmission of light through the filtrate specimen shall be measured by means of a photo-electric colimeter, turbimeter, or equivalent method. The result shall be expressed as the equivalent of parts per million of silica (diatomaceous earth or fuller's earth) in water as described in applicable paragraphs of Standard Methods for the Examination of Water, Sewage, and Industrial Waste (see 2.2).

4.6 Examination of preparation for delivery requirements.

4.6.1 Inspection of fabricated bags and bale-boards. Fabricated bags and bale-boards shall be inspected to determine conformance with the applicable requirements of 5.1 and 5.2. The acceptable quality level shall be 4.0 defects per 100 units and the inspection level shall be S-2 of MIL-STD-105.

4.6.2 Examination of the end item. The end item (completely packaged and packed feathers) shall be inspected in accordance with the examinations and tests specified in 4.6.2.1, 4.6.2.2, and 4.6.2.3.

4.6.2.1 Weight. The lot shall be unacceptable if the total net weight of the bags in the sample is less than the total net weight indicated on the bags. The number of bags to be selected for the weight examination shall be in accordance with S-4 of MIL-STD-105. The net weight of the material contained in the sample bags shall be obtained by deducting the tare weight from the gross weight of the sample bags.

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4.6.2.1.1 Standard weight basis. When specified, or in case of dispute, the weight basis for acceptance shall be determined by correction of unconditioned weight, to weight at equilibrium under standard atmospheric conditions as defined in Fed. Std. No. 148. At the time of weighing, a random sample of product not less than 1/4 pound shall be taken from the contents of the bag, and placed immediately in a tared closable container. The sample shall then be weighed in the closed container. The sample shall then be spread in a shallow layer in a tray and stored in the standard atmosphere for not less than 24 hours, with occasional stirring. The conditioned sample shall then be weighed, and the gain or loss calculated to a percentage based on the original sample weight. The percentage correction shall then be applied to the unconditioned weight of the contents of the bag to determine its weight on the standard weight basis.

4.6.2.2 Packaging examination. The sample unit shall be one bag of feathers. The inspection level shall be S-1 of MIL-STD-105 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below:

Defects

Not packaged in accordance with specified packaging level.
Bag not new, or any patch.
Not closed as specified.
Wire tie broken or not tightly turned to accomplish a secure closure.
Any hole, rip, tear, open seam; any weak places in bag which may develop into a hole; backtacking and backstitching not as specified.
Bag wet.

4.6.2.3 Packing examination. The sample unit shall be one bale or shipping container. The inspection level shall be Level I of MIL-STD-105 and the acceptable quality level shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below:

Defects

Not packed in accordance with specified packing level.
Bags not compressed to specified size.
Does not contain the specified number of bags.
Any board missing or strap missing.
Baling board or strap too loose or not placed to assure safe delivery of contents.
Marking omitted, incomplete, illegible, incorrect, not as specified or not in proper location.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Fifty ± 5 pounds of feathers of one type only shall be packaged in a new (not used or laundered) plain textile bag fabricated from cotton sheeting conforming to class 1, type VIII of CCC-C-432, or osnaburg conforming to class 2 of CCC-C-429. Cloths of equivalent weight and construction will be acceptable.

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5.1.1.1 Bag, construction. The bags shall have a finished length and width, measured flat, of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The side and bottom shall be seamed with 5 to 6 stitches per inch in seam type SSa-2, with either stitching type 401 or 301, conforming to Fed. Std. No. 751. The stitching shall be at least 5/8 inch from the cut edge of the cloth or 1/4 inch from a selvage edge and the rows shall be at least 1/8 inch apart. The ends of all stitching, when not caught in other seams or stitching, shall be backstitched or backtacked at least 1 inch. The thread used for seaming shall be cotton, machine type IA1 or IB1, unbleached, ticket No. 12, 4-ply conforming to V-T-276.

5.1.1.1.1 Type 301 stitching. Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.1.2 Type 401 stitching. Thread breaks in stitching shall be over-stitched not less than 1-1/2 inches at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.2 Closure. The closure of the top of the bag after filling shall be effected by means of two double loops of 0.072 inch thick galvanized soft iron or steel wire ties of not less than 6 inches long, having a 1/2 inch diameter eye formed at each end. One tie shall be applied at the base of an ear formed by gathering the cloth evenly and the second tie applied to a distance approximately 1 inch from the first tie with the twisted ends of two ties alternated and facing in opposite direction.

5.1.2 Level C. Feathers shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Three bags of feathers of one type only shall be placed between two baling boards and compressed by machine so as to form a bale approximately one-fifth of the original volume. Bales thus formed shall measure approximately 60 by 24 by 20 inches.

5.2.1.1 Baling boards. Baling boards shall be solid fiberboard conforming to type SF, class weather-resistant, grade V4s of PPF-F-320. Two such baling boards, size 58 by 36 inches, each with a score line placed 6 inches from each 58 inch edge, shall be used in each bale; one board on top of bags being baled, and one on the bottom. The boards shall be bent at the score lines thereby forming a fiberboard cap 58 inches in length, 24 inches in width, with 6 inch flanges extended down over the side of the bale.

5.2.1.2 Strapping. Not less than eight straps shall be used on each bale; straps shall be affixed girthwise around the sides, top, and bottom of the bale over the baling boards. The two outside straps shall be placed 4 inches in from each end and the intermediate straps placed equidistant between the end bands. All strapping shall be not less than 3/8 inch wide by 0.020 inches thick conforming to type I, class A of QQ-S-781 or type II of PPP-S-760.

5.2.2 Level C. Feathers packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The processed landfowl feathers covered by this specification are intended for the following uses:

Types I and III - Filling materials in pillows and other items.
Type II - Specialized filtration applications.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of feathers required (see 1.2).
(c) First article. When preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Processing. With reference to the processing requirements of 3.3, the following procedure is suggested to yield a satisfactory product:

(a) Within 30 minutes after wet plucking, the feathers are given a clear water rinse at temperature not over 105°F.
(b) Rinsed feathers are then agitated for 20 minutes in water at 95° ± 5°F, to which has been added 1 ounce of blood-solubilizing agent for each 10 gallons of water.
(c) Feathers are then scoured with detergent in warm water, and thoroughly rinsed. For types I and III feathers, in the final rinse a commercial sour of prescribed type is added.
(d) Feathers are then centrifuged, and dried in a steam heated dryer with means of agitation. The drying cycle is controlled to attain moist heat of 165°F, or higher at the exhaust stack for a minimum of 5 minutes to effect sterilizing conditions.

6.4 Comparison of types as pertain to C-F-158 and MIL-F-11836 is as follows:

MIL-F-11836B	C-F-158C
Type I	Type I
Type II	Type II
C-F-158b	C-F-158C
(One type)	Type III

MILITARY CUSTODIANS:

Army - GL
Air Force - 82

Review activities:

Army - MD
Air Force - 82

Preparing activity:

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